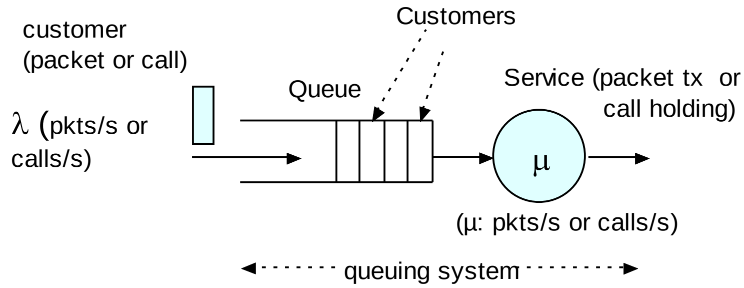
Queuing Theory

Summary of Main Formulae

- Basic Structure of a Queuing System
- *M/M/1* Queuing System
- Little's Theorem

Basic Structure of a Queuing System

- Customers (= packets) arrive at random times to obtain service
- Service time (= transmission delay) is *L/C*
  - *L*: mean Packet length in bits (Packet Size, PKTSIZE)
  - *C*: Link transmission capacity in bits/sec



- Assume that we already know:
  - Customer arrival rate (customers/sec, packets/sec, frames/sec)
  - Customer service rate (customers/sec, packets/sec, frames/sec)

- We want to know:
  - Mean number of customers in the system (N)
  - Mean delay per customer (T or RT)

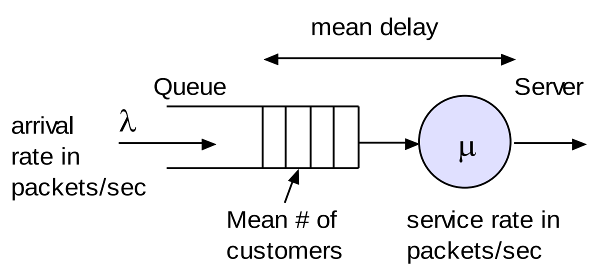


Fig. Basic structure of a queuing system

Arrival Pattern

- Customers are drawn from a *population* or *input source*.
- Population size may be *infinite* or *finite.*
- Statistical pattern by which arriving customers are generated is known as the *arrival pattern*
- The simplest and most useful model is the completely random arrival process: i.e. the *Poisson process.*

Service Distribution and Capacity

- The amount of service required by a customer is called the *service demand* or *work*.
- The server is usually a CPU or a transmission line, while the customers are messages or data (e.g. a call or a packet). In this case the unit of work is in bits or bytes.
- Generally it is assumed that the population is homogeneous in which case service demands are identically distributed with a common distribution called the *service distribution*.
- In more complex systems, the customers may be classified into different types each with its own distribution of service demand.

Service Capacity and Rate

- *Processing rate* or the *capacity (C)* of the server must also be specified.
- The units of *C* depends on the type of service: if the server is a CPU its unit is MIPS or if it is a transmission channel its unit is bps (representing the transmission rate).
- Assume that the work demanded by the customer is S (service units) and the server has capacity C (service units/sec), then:
- T_s= *Service time*= E[S]/C in seconds
- $\mu = 1/T_s = C/E[S]$ is the completion rate of the *service rate.*

The Scheduling Discipline

- Need to consider how the customers are scheduled and served. Some of the well known disciplines are:
- FCFS: First come first served discipline
- LCFS: last come first served

- If the system differentiates between customers according to their importance then the procedure is called the priority scheduling discipline. The queuing system is known as a *priority queue*.
- Priority queue disciplines are of two types: *preemptive* and *non-preemptive*.

Uses of Queuing Theory?

- Theoretical framework for analyzing network performance indices such as delay and throughput. It also allows calculation of buffer requirements as well as resource dimensioning
- Often requires simplifying assumptions since realistic assumptions make meaningful analysis extremely difficult
- Provide a basis for adequate delay approximation, estimation of system performance under different conditions.

Delay in PS Networks

- PS networks consist of a number of packet switches (e.g. routers/gateways) which are connected by transmission lines
- Each of these units introduce delay during the packet travel from source to destination.
- Delay in packet switches include:
  - **Processing delay** (incl. interrupt service, processor scheduling, routing decisions and other processing delays, buffer copying, internal bus access delays etc.)
  - **Queuing delay**
- Delay through transmission lines include:
  - **Transmission delay**
  - **Propagation delay**

End-to-End (Packet) Delay

- E-E (packet) delay is the sum of delays on each subnet link traversed by the packet
- Total link delay consists of the sum of:
- Processing, Queuing, Transmission, and Propagation delays.

A Switching Node Model

Processing Delay

- Delay between the time the packet is correctly received at the switching node of the link and the time the packet is assigned to an outgoing link queue for transmission

Queuing Delay

- Delay between the time the packet is assigned to a queue for transmission and the time it starts transmission by the transmitter (server) at the SRC node.

Transmission Delay

- Delay between the times of transmitting the first and the last bits of the packet from the SRC node

Propagation Delay

- Delay between the time the last bit is transmitted at the SRC node of the link and the time the last bit is received at the DST node

Queuing System (1)

- Customers (= packets) arrive at random times to obtain service
- Service time (= transmission delay) is L/C
  - L: Packet length in bits
  - C: Link transmission capacity in bits/sec

Queuing System (2)

- Assume that we already know:
  - Customer arrival rate
  - Customer service rate
- We want to know:
  - Mean number of customers in the system
  - Mean delay per customer

Definition of Symbols (1)

- p_n = Steady-state probability of having n customers in the system
- λ = Arrival rate (inverse of mean interarrival time). Units: *customers/sec.*
- μ = Service rate (inverse of mean service time). Units: *customers/sec.*
- N = Mean number of customers in the system

Definition of Symbols (2)

- Q = Mean no. of customers waiting in queue
- T = Mean customer time in the system=RT (response time)
- $W = T_w$ = Mean customer waiting time in queue (does not include service time)
- T_s = $1/\mu$= Mean service time
- X_2 = Second moment of service time

Little's Theorem

N = Mean number of customers
λ = Arrival rate (customers/s)
T = Mean customer time in the system

$$N = \lambda T$$

- Holds true for almost every queuing system that reaches a steady-state
- Express the natural idea that crowded systems (large N) are associated with long customer delays (large T) and reversely
    - Application of Little's Theorem (1)
- Q = Mean # of customers waiting in queue
- W = Mean customer waiting time in queue

$$Q = \lambda W$$

- T_s = Mean service time

  ρ = Mean # of packets under transmission

$$\rho = \lambda T_s$$

ρ is called the utilization factor (= the proportion of time that the line is busy transmitting a packet or *load*)

M/M/1 Queuing System: Results (1)

- Utilization factor (proportion of time the server is busy)

$$\rho = \frac{\lambda}{\mu}$$

- Mean interarrival time, IAT :

$$IAT = \frac{1}{\lambda}$$

- Mean Service Time, T_s:

$$T_s = \frac{1}{\mu}$$

- Probability of *n* customers in the system

$$p_n = \rho^n (1-\rho)$$

- Mean number of customers in the system

$$N = \frac{\rho}{1-\rho}$$

M/M/1 Queuing System: Results (2)

- Mean customer time in the system, *T* or *RT* :

$$T = RT = \frac{N}{\lambda} = \frac{1}{\mu - \lambda}, \quad RT: \text{ResponseTime}$$

- Mean number of customers in queue:

$$Q = \lambda W = \frac{\rho^2}{1-\rho}$$

- Mean waiting time in queue, (W, W_q, T_w, Tw_q) :

$$W = T - \frac{1}{\mu} = \frac{\rho}{\mu - \lambda}$$

Sample Problem

Given: ISDN B-ch is used to transmit frames which have negative exponential distribution with mean 1000 Bytes and the mean frame arrival rate from a Poisson source is 5 packets/sec.

i) Calculate mean number in the system
ii) Find Response Time (Sojourn Time)
iii) Find the probability that the server is idle.
iv) Find mean waiting time in the queue.

Solution

- λ = 5 fps, C = 64,000 bps
- 1/u = Mean Frame Size = 1000B = 8000 bits/frame
- μ:= mean service rate = uC = 64,000/8,000 = 8 fps.
- $\rho = \lambda / \mu$ and $N = \rho / (1-\rho)$,

Hence,

- ρ = 5/8 = 0.625 N = 0.625/(1-0.625) = 0.625/0.375= 1.666
- RT=N/ λ=1.666/5 = 0.333 s= 333 ms.
- Pr{0} = 1- ρ = 1 - 0.625 = 0.375
- W_q=Mean Waiting Time in Queue= RT-T_s=333-125 ms = 208 ms.